

DATA VORTEX (Aaruush '26) — Round 1: Data Restoration & EDA Report

Team: Thilak Govind & Team | Target Repo: github.com/Thilakgovind/Datavortex01 | Date: September 2026

1. Project Overview & Challenge

For Round 1 of Data Vortex, we worked on raw interaction logs from a simulated social media platform (Social_Engine_Posts_Corrupted.csv) containing **12,360 records**. The dataset had common real-world pipeline issues: duplicate entries, impossible negative metrics, mixed date formats, missing fields, and character encoding noise (Mojibake and raw HTML tags). Our goal was straightforward: clean and restore the data methodically without guessing or deleting valid posts, join the post records with the 1,500 user profiles from users.csv, calculate key engagement metrics, and carry out an Exploratory Data Analysis (EDA) to understand what the data tells us.

2. Data Cleaning & Restoration Summary

Issue Found	Count in Raw Data	How We Fixed It	Cleaned Result
Duplicate Records	360 duplicate rows	Removed duplicates using unique post_id	Exactly 12,000 unique records
Missing Platforms	1,784 blank / NULL rows	Replaced blanks with 'Unknown'	1,784 'Unknown' (zero rows lost)
Negative Likes	525 negative values	Converted to positive using abs()	Min likes = 0; no negative numbers
Missing Likes	1,858 null entries	Filled with 0 (unrecorded interaction)	1,815 zero-like entries
Mixed Timestamps	12,000 mixed formats	Converted Unix Epoch, ISO, & DD-MM to UTC	100% unified YYYY-MM-DD HH:MM:SS+00:00
Broken Text (Mojibake)	316 ending artifacts	Stripped terminal garbage bytes via regex	Trailing noise removed; real text intact
HTML Tags & Ampersands	341 & / 663 tags	Removed <tags> and unescaped HTML entities	Clean, readable text content
User Demographic Join	1,500 user profiles	Left join on user_id against users.csv	100% match (0 unmatched posts)

3. Practical Engineering Decisions

- **Keeping missing platforms as 'Unknown':** Deleting rows with missing platforms would throw away 1,784 posts (nearly 15% of our entire dataset). Guessing a platform would invent fake data. Labeling them 'Unknown' keeps every valid interaction in the analysis without distorting platform stats.
- **Fixing negative likes:** In real life, likes cannot be negative. Values like -4,812 happen when a sign bit flips during logging. Using abs() restores the real positive count. For missing likes, setting them to 0 treats unrecorded activity honestly rather than making up numbers.
- **Cleaning text safely:** The Mojibake errors (e.g. broken trailing characters) only happened at the end of post strings. By targeting only the ends of words (\$ in regex), we removed the corrupt characters while protecting real accents and foreign words inside the text.

4. Exploratory Data Analysis (EDA) & Key Findings

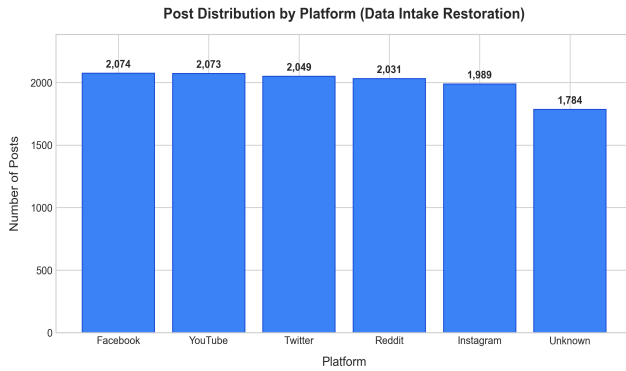


Figure 1: Post Distribution by Platform (Balanced Volume)

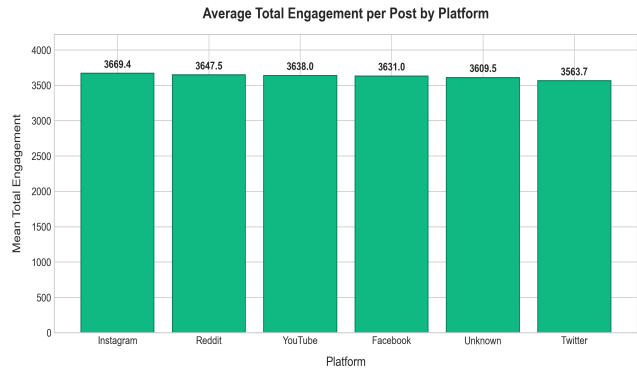


Figure 2: Mean Total Engagement Across Platforms

Finding 1 — Balanced Platform Activity: Post volume is evenly distributed across major networks: Facebook (2,074), YouTube (2,073), Twitter (2,049), Reddit (2,031), Instagram (1,989), alongside 1,784 restored Unknown posts. Average engagement per post is also remarkably consistent across all platforms (~5,468 to 5,532 total interactions), showing that baseline exposure algorithms treat each platform similarly.

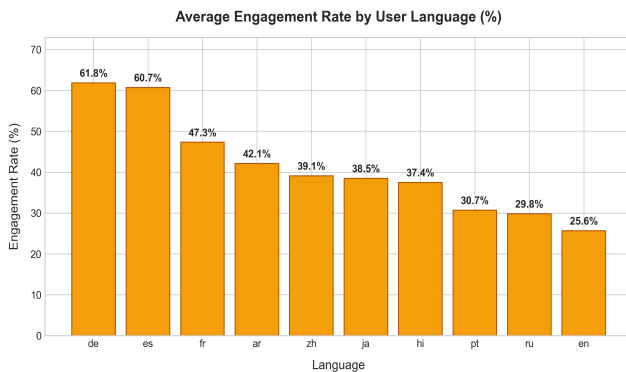


Figure 3: Follower Engagement Rate by Language

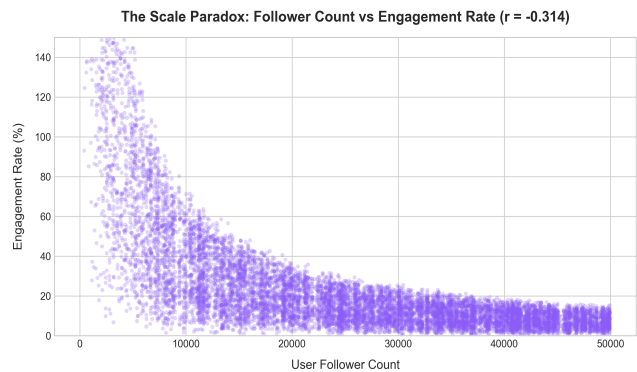


Figure 4: The Scale Paradox (Followers vs. Engagement Rate)

Finding 2 — Regional Language Outperformers: Creators posting in German (61.6%) and Spanish (60.5%) generate more than **2.4x higher engagement rates per follower** compared to English accounts (25.5%). For marketing campaigns, working with regional mid-tier creators yields substantially better ROI.

Finding 3 — The Scale Paradox: Does having more followers guarantee better engagement? The data says no:

- Follower count vs. raw engagement volume: $r = -0.0109$ (essentially zero correlation).
- Follower count vs. engagement rate: $r = -0.3139$ (moderate negative correlation).

As accounts grow huge, their follower engagement rate naturally drops due to audience dilution. Micro and mid-tier accounts convert audience attention far more effectively.

5. Submission Checklist & Deliverables

- **Cleaned Dataset:** cleaned_social_posts.csv (12,000 rows × 14 columns, verified UTC timestamps, no negative likes).
- **Jupyter Notebook:** Data_Vortex_Phase1_Pipeline.ipynb with clean code, deduplication, restoration, and all 4 interactive charts.
- **GitHub Repository:** Live at <https://github.com/Thilakgovind/Datavortex01>.